Chapter 1: Heredity and Evolution

Q1. Explain morphological evidence of evolution with suitable examples. (4 Marks)
Subtopic: Morphological evidences
Morphological evidence is based on similarities and differences in external body structures of organisms.
Similarities in body form, shape and structure indicate common ancestry and evolutionary relationship.
Example 1: Limbs of different vertebrates show similar arrangement of parts (upper arm, forearm, hand) though their functions differ.
Example 2: Similar body plan and arrangement of organs among related organisms suggests descent from a common ancestor.
Important points for full marks:
Write definition + at least two example-based explanations.
Link the similarity to common ancestry and evolution.

Q2. Explain anatomical evidence of evolution with suitable examples. (4 Marks)
Subtopic: Anatomical evidences
Anatomical evidence is based on similarities and differences in internal structures and organs of organisms.
Similar internal organ arrangement indicates that organisms may have evolved from common ancestors.
Example 1: Similar skeletal plan in vertebrates supports evolutionary relationship.
Example 2: Similar structure of heart, brain and other organs in vertebrates indicates common origin.
Important points for full marks:
Mention internal structure comparison clearly.
Give at least one skeletal or organ-based example.

Q3. What are vestigial organs? Give any four examples of vestigial organs in human beings. (3 Marks)
Subtopic: Vestigial organs
Vestigial organs are reduced and functionless organs present in an organism.
They were functional in ancestors but have lost their function during evolution.
Examples in humans (any four): appendix, tail bone (coccyx), wisdom teeth, muscles of pinna (ear muscles), body hair.
Important points for full marks:
Write definition + at least four correct examples.

Q4. Explain palaeontological evidence of evolution. (4 Marks)
Subtopic: Palaeontological evidences
Palaeontological evidence is obtained from fossils preserved in rocks.
Fossils provide information about organisms that lived in the past and show changes over long periods of time.
By studying fossils from different rock layers, we observe gradual changes and appearance of new forms of life.
Thus fossils establish the sequence of evolution and confirm that present organisms evolved from earlier forms.
Important points for full marks:
Write the keyword fossils and rock layers/time sequence.
Explain how fossils show gradual change.

Q5. What is a connecting link? Explain with one suitable example. (3 Marks)
Subtopic: Connecting links
A connecting link is an organism that shows characteristics of two different groups.
It forms a bridge between the two groups and indicates evolutionary relationship.
Example: Duck-billed platypus shows reptilian character (egg laying) and mammalian character (mammary glands).

Q6. Explain embryological evidence of evolution. (4 Marks)
Subtopic: Embryological evidences
Embryological evidence is based on similarities in early developmental stages of embryos of different organisms.
Early embryos of many vertebrates show common features such as gill slits and a tail.
These similarities indicate that these organisms share common ancestry.
Therefore embryology supports the theory of evolution.
Important points for full marks:
Write gill slits and tail as common embryonic features.
Conclude with common ancestry.

Q7. Explain Darwin's theory of natural selection. (5 Marks)
Subtopic: Darwin's theory of natural selection
Darwin proposed that evolution occurs by natural selection.
Organisms produce more offspring than can survive, leading to struggle for existence.
Individuals show variations; some variations are favourable and heritable.
Those with favourable variations survive and reproduce more, while others are eliminated.
Over many generations, favourable variations accumulate and may result in formation of new species.
Important points for full marks:
Use the sequence: overproduction, struggle, variation, survival and accumulation.

Write the phrase favourable variations and heritable.

Q8. State Lamarckism and explain it with suitable points. (4 Marks)
Subtopic: Lamarckism
Lamarckism states that organisms acquire certain characters during their lifetime due to changes in environment and use and disuse of organs.
According to Lamarck, these acquired characters are inherited by the next generation, leading to evolution.
Main points: effect of environment, new needs, use and disuse of organs and inheritance of acquired characters.
Example often explained: development of longer neck in giraffe due to continuous stretching to reach leaves.
Important points for full marks:
Write the terms use and disuse and inheritance of acquired characters.
Give one example (giraffe) clearly.

Q9. What is speciation? Explain the role of isolation in speciation. (4 Marks)
Subtopic: Speciation
Speciation is the process of formation of a new species from an existing species.
Isolation means separation of a population into groups due to geographical, ecological or reproductive barriers.
Due to isolation, gene flow stops between the groups and variations accumulate separately in each group.
After a long period, the groups become so different that they cannot interbreed, and a new species is formed.
Important points for full marks:
Write the keywords isolation, no gene flow and cannot interbreed.
Link isolation to accumulation of variations.

Q10. Write a short note on human evolution. (5 Marks)
Subtopic: Human evolution
Human evolution explains the gradual development of modern humans from earlier ancestors over a long time.
Early human ancestors evolved from primate-like forms and showed progressive changes in body and brain.
Important changes include upright posture, bipedal locomotion, increase in brain size and development of speech and intelligence.
Use of tools, control over fire and social living further supported human progress.
Thus modern humans evolved through gradual changes over generations.
Important points for full marks:
Write bipedal locomotion and increase in brain size.
Mention tool use and social living for completeness.